Thinking Out Loud, and What It Costs

ME/CE/AM 295

Agentic AI in Scientific Discovery
9 units (3-5-1)

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Lecture 4

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“Reasoning” You Can Audit



From Language Models to Autonomous Agents

WHERE WE LEFT OFF · WEDNESDAY

The schema declares a shape. The model can still be wrong inside it: with no tool, its arithmetic is pattern-matching. Today: what the model can do by itself before it answers, what that is worth, and what it costs.

PART 1
The Scratchpad

—

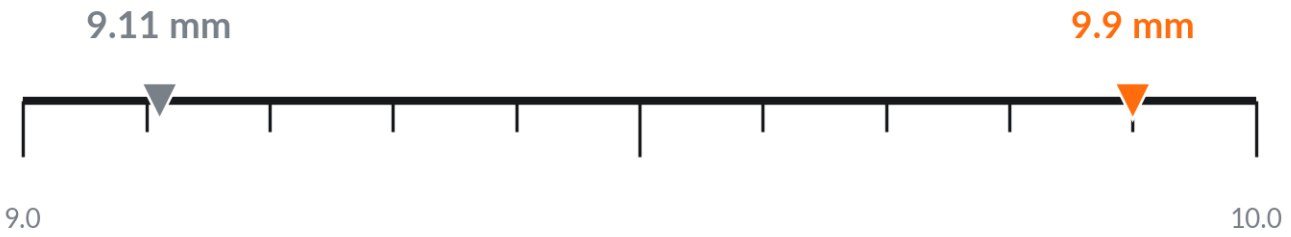
PART 2
The Price

one pass → many passes → a vote → a tree → the bill → the reasoning field
(no tools yet — that is next Wednesday)

(2024) Which Plate Is Thicker?

Two plates. Hands up: 9.11 mm or 9.9 mm? Then the same question to a model, asked two ways.

"Which plate is thicker?" — the answer-at-once model says 9.11, because 11 > 9



Reasoning is critical to get the right answer (otherwise it's just based on pattern completion)



ASKED: "Reply with the number only."

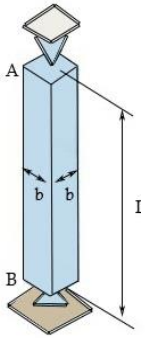
"9.11 mm — since 11 > 9."

ASKED: "Write out your reasoning, then give the number."

"Initial instinct says 9.11 since 11 > 9… wait, let me break down decimal places. 9.11 has 1 in the tenths place, 9.9 has 9. 9.9 is larger. Rewriting final answer."

The 9.11 vs 9.9 failure is real and widely reproduced on 2024-era models; **current models mostly get it right — because they write first.**

Schema-Valid, Wrong Inside — the Hinge



$$P_{cr} = \frac{\pi^2 EI}{(KL)^2}$$

Same data, same model. The only difference is the instruction: answer immediately, or write the intermediate steps out first.

`E = 200 GPa,  I = 4.5×10⁶ mm⁴,  L = 3.2 m.  →  {critical_load_kN: float}`

A: “Answer immediately. No working.”

B: “Intermediate quantities with units, then the result.”

A · ANSWER AT ONCE — one read, then digits

1 240 kN

✗ valid JSON, in range — wrong by 43 %

The digits were pattern-completed, not computed.

B · STEPS FIRST — each line is read by the next pass

```
π²EI   = 8.88×10¹² N·mm²
(KL)²  = 1.024×10⁷ mm²
```

867 kN

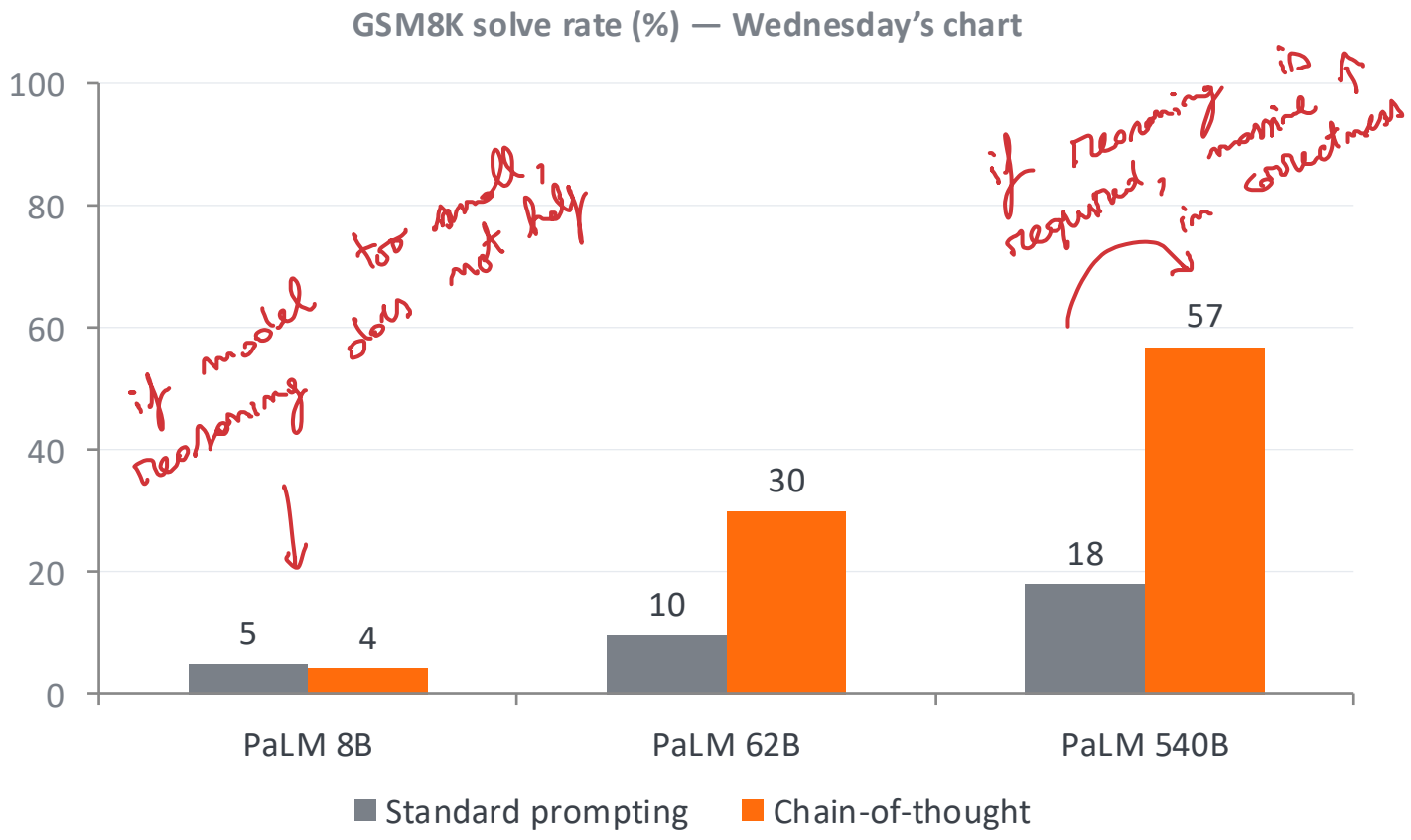
✓ the division acts on written numbers

Nothing is remembered between passes except what is written. Written steps are the model’s only working memory.

The step of writing reasoning is what allocates memory to get the right answer

Chain-of-Thought (2022) — Show a Worked Example

Wei et al.: put worked rationales in the few-shot examples and the model writes its own before answering. Same weights, more passes.



WHAT CHANGED

Nothing in the model. The prompt gives it a pattern for spending passes.

THE CATCH

8B gains nothing; 62B triples. A scratchpad helps only where the model can use it.

THE PRICE

Eight worked examples ≈ 1 000 input tokens on every call, plus a rationale on every output.

Wei, J., Wang, X., Schuurmans, D., et al., NeurIPS 35 (2022), arXiv:2201.11903 · GSM8K values from Table 2 (8.5 K grade-school arithmetic word problems)

Zero-Shot Chain-of-Thought — Six Tokens, Not a Thousand

Kojima et al.: append “Let’s think step by step” and the model writes a rationale with no examples at all.

Benchmark	Zero-shot	Zero-shot CoT
MultiArith	17.7 %	**78.7 %**
GSM8K	10.4 %	**40.7 %**

WHY FIVE WORDS WORK

The phrase is appended as the first assistant tokens. The next pass reads it, and the most probable continuation of “Let’s think step by step” is a step.

The scratchpad is the model’s own output, read back

The 5 words force the model to re-read their own out-put at every iteration, therefore better accuracy

THE LADDER, APPLIED TO REASONING

1 **DESCRIBE IT — zero-shot CoT**

“Show the intermediate quantities with units, then the result.” About ten tokens.

2 **SHOW IT — few-shot CoT**

Worked examples in the prompt. Pins the style of reasoning; ~1 000 tokens per call.

3 **DECLARE IT — a reasoning field**

A string field, first in the schema. Logged, bounded, validated.

pre-encoded in all new models now

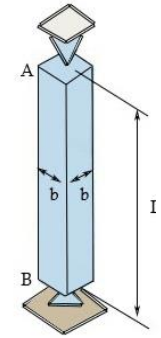
embedding the CoT in the agent

Engineering phrasing beats magic words: “state each intermediate quantity with its unit before the result” says what a step is.

Kojima, T., Gu, S. S., Reid, M., Matsuo, Y., Iwasawa, Y., NeurIPS 35 (2022), arXiv:2205.11916 · text-davinci-002 (a 2022 GPT-3 model), Table 1; MultiArith = multi-step arithmetic word problems · Optional reading

Sign Off or Send Back — Trace 1

The apprentice at the whiteboard. Pin-ended column problem. Vote!



$$P_{cr} = \frac{\pi^2 EI}{(KL)^2}$$



THE TASK The Euler column from the hinge slide, asked the B way: “state each intermediate quantity with its unit, then the result.” No tools — the apprentice alone.

“E = 200 GPa, I = 4.5 × 10⁶ mm⁴, L = 3.2 m.
π²EI = 9.87 × 200 × 4.5 × 10⁶ = 8.88 × 10⁹.
L² = 10.24.

P = 8.67 × 10⁸ N = 867 000 kN. That is a very stiff column, which makes sense for steel.”

SIGN OFF

SEND BACK — which line?

Sign Off or Send Back — Trace 2

Right number. Would you accept this from a human colleague?



THE TASK The Euler column from the hinge slide, asked the B way: “state each intermediate quantity with its unit, then the result.” No tools — the apprentice alone.

“A pin-ended steel column of this size typically carries around 800–900 kN. I’ll say 867 kN.”

SIGN OFF

SEND BACK – which line?

Sign Off or Send Back — Trace 3

Watch for one word.



THE TASK The Euler column from the hinge slide, asked the B way: “state each intermediate quantity with its unit, then the result.” No tools — the apprentice alone.

“$(KL)^2 = 3.2^2 = 10.24\ m^2$… wait — I have I in mm^4 and E in N/mm^2, so L must be in mm: $3\,200^2 = 1.024 \times 10^7\ mm^2$.
Recomputing: $P = 8.88 \times 10^{12} / 1.024 \times 10^7 = 8.67 \times 10^5\ N = 867\ kN$.”

SIGN OFF

SEND BACK — which line?

Sign Off or Send Back — Trace 4

An FEA run. The solver has failed once; the apprentice is fixing it. Every step follows from the one before.



THE TASK Run the model; if it fails, diagnose and fix, then re-run. The solver's message is pasted back after each attempt.

THE SOLVER SAID *singular stiffness matrix* — at least three ordinary causes: under-restraint, a disconnected node, a zero-stiffness rotation.

"The stiffness matrix is singular, so a support must be missing. Adding a pin at node 1.

Still singular — add a roller at node 2.

Still singular — fix node 3 fully.

Still singular — the mesh must be wrong; regenerate it."

SIGN OFF

SEND BACK — which line?

The Verdicts — and What Each One Teaches

Four traces, four different ways a written rationale can mislead or save you.

Trace	Verdict	The line	The lesson
1 · Confident unit slip	**Send back**	"…which makes sense for steel."	A narrative written after the number. Read traces for numbers and units, not reassurance.
2 · Right by pattern	**Send back**	There is none — nothing was derived	A correct number you cannot check is not a result. Ask for the intermediate quantities.
3 · The backtrack	**Sign off**	"wait —"	A written step gave a later pass something to catch. This is what you are buying with passes.
4 · Entrenchment	**Send back**	Line 1: the wrong diagnosis	Every later pass read it as fact. Self-written diagnoses need an external check — Rule 5, and Wednesday.

All models think before responding. Effort changes how many steps it does

The Scratchpad Moved Inside the Model — Thinking in 2026

Current models write a hidden rationale before the answer on their own. You steer how much with effort; you pay for it as output tokens.

```
from anthropic import Anthropic
client = Anthropic()

resp = client.messages.create(
    model=MODEL,                    # pinned, logged
    max_tokens=4096,                # thinking + answer
    thinking={"type": "adaptive"},
    output_config={"effort": "high"},
    system=SYSTEM,
    messages=[{"role": "user",
               "content": prompt}],
)
for b in resp.content:
    if b.type == "thinking": summary = b.thinking
    if b.type == "text":     answer = b.text
resp.usage.output_tokens_details.thinking_tokens
```

THREE WAYS TO GET A RATIONALE

1. Prompted (CoT slides): visible, yours to read.
2. Declared: a field in your log.
3. Native: hidden, summarised, billed in full.

WHAT EFFORT IS

Soft guidance on how readily and deeply the model thinks: low → max. Default high (medium on Opus 5.5, Haiku 5.5). Not a token count — max_tokens is the hard cap.

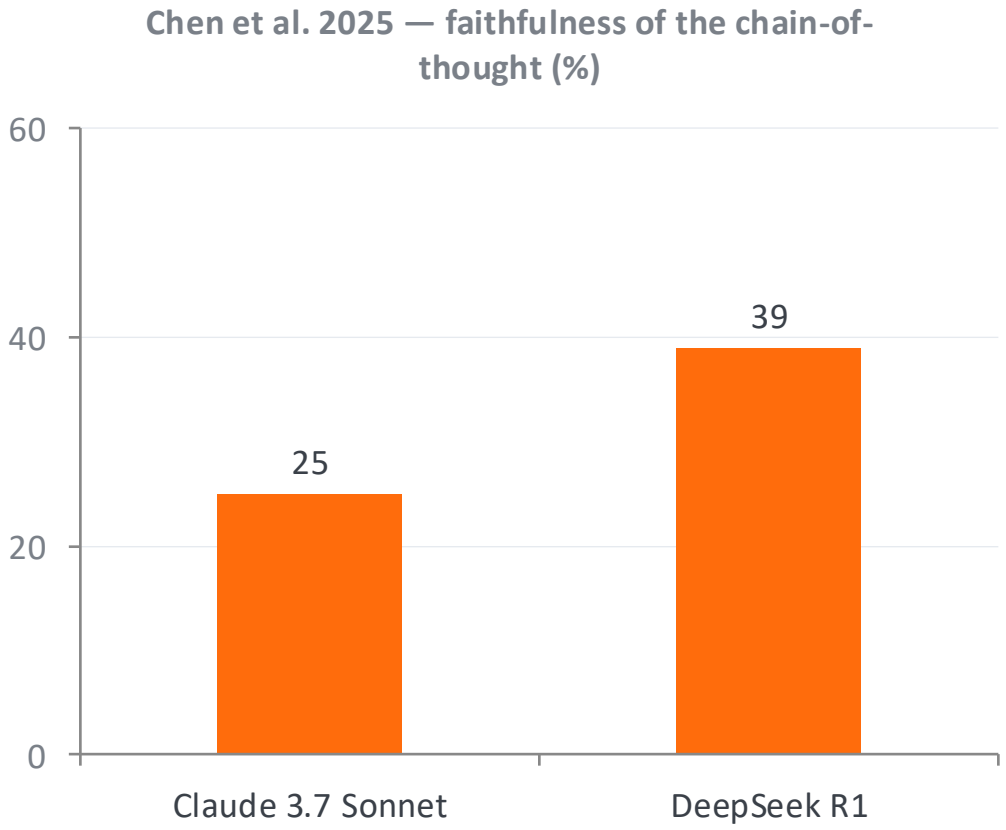
WHAT IT IS NOT

Not a determinism knob (T is still not yours) and not evidence (next slide). Changing effort mid-conversation invalidates the prompt cache.

Anthropic, Thinking / Steering thinking / Effort guides, platform.claude.com/docs (checked Oct 2026) · Required reading

The Trace Is a Narrative, Not a Proof

A rationale is written by the instrument whose answer it explains. Two experiments measure how often it tells you what actually drove the answer.



model lied in reasoning

TURPIN ET AL. 2023

Bias the few-shot examples so the answer is always (A). Accuracy drops by up to 36 points — and the written rationale never mentions the pattern. It argues for (A) on the merits.

CHEN ET AL. 2025

Plant a hint in the prompt. When the model's answer follows the hint, its thinking admits it 25 % (Claude 3.7 Sonnet) and 39 % (R1) of the time; less on harder questions.

ENGINEERING IMPLICATION

Read a trace for its numbers and units, the way you read a colleague's hand calc — not for reassurance. The thesis at the scale of one rationale: the instrument cannot be its own evidence.

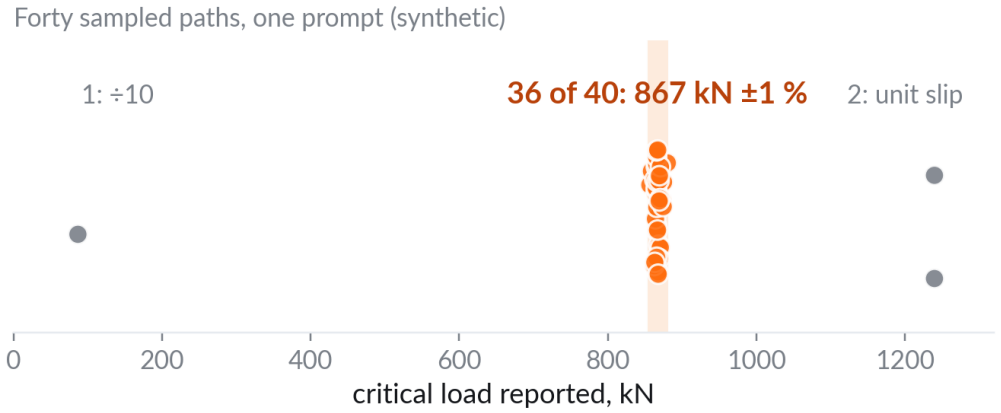
Turpin, M., et al., NeurIPS 36 (2023), arXiv:2305.04388 · Chen, Y., et al. (Anthropic), 2025, arXiv:2505.05410 · Recommended reading (Chen et al.)

Self-Consistency — Sample Many Paths, Vote on the Answer

Wang et al.: run the same chain-of-thought prompt N times at T > 0 and take the majority answer. Different paths, same destination, is evidence.

56.5 % → 74.4 %

GSM8K, PaLM-540B: one greedy chain versus a vote over 40 sampled chains; most of the gain arrives by 10–20 paths. Below: which dot is the most useful one?



NEEDS T > 0 — OR DEFAULT SAMPLING

Identical paths give a vote that proves nothing.

On models with no T knob, default sampling already varies; confirm the paths differ before you count.

NEEDS A COMPARABLE ANSWER

A number with a tolerance, a section designation, a class label. For prose there is nothing to vote on — and a schema (Wednesday) is what makes the answers comparable.

COSTS N× — AND IT IS A MEASUREMENT

N separate API calls, N rationales, N bills. Like repeating a measurement: it beats random error, not systematic — forty paths with the same wrong formula agree.

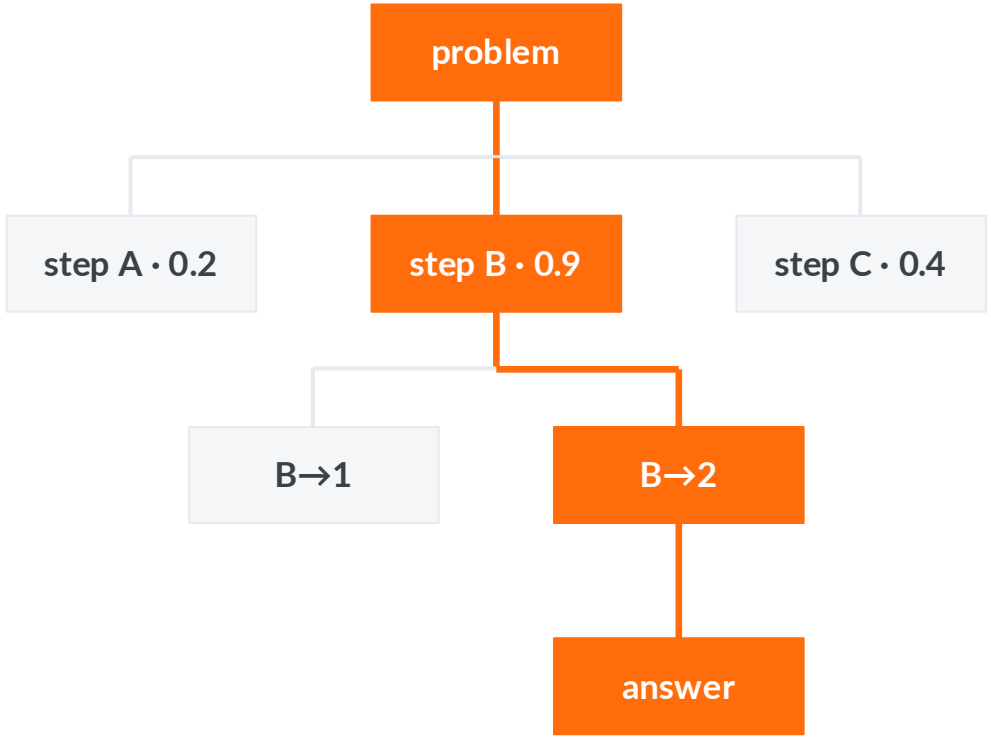
Wang, X., Wei, J., Schuurmans, D., et al., ICLR (2023), arXiv:2203.11171 · PaLM-540B, 8-shot CoT, 40 sampled paths, majority vote · 56.5 % is Wang et al.'s own greedy baseline; Wei et al. report 56.9 % for the same set-up · Recommended reading

Evaluates probability of each answer and choses the highest probability one (very expensive in tokens)

Tree of Thoughts — Branch, Score, Prune, Backtrack

↳ need to be prompted

Yao et al.: generate several candidate next steps, evaluate each, expand the best, abandon dead ends.
A search, with the model as both proposer and judge.



grey branches are scored and dropped; the orange path is the only one taken to the end

4 % → 74 %

Game of 24, GPT-4: chain-of-thought 4.0 %, CoT + 100-path vote 9.0 %, tree search (breadth 5) 74 %.

WHEN IT PAYS

→ useful for when the answer can be scored

When steps can be scored before the end — a partial design that already violates a constraint is pruned. Branch and bound, with the model proposing branches.

WHAT IT COSTS, AND WHAT IT PREVIEWS

Breadth × depth × passes per node — orders of magnitude more tokens than one chain. Make the judge a tool and you have an agent's planner (Wednesday).

Yao, S., Yu, D., Zhao, J., et al., NeurIPS 36 (2023), arXiv:2305.10601 · Game of 24 (reach 24 from four numbers with + − × ÷), GPT-4, Table 2 · Optional reading

Four Ways to Spend Passes

Each rung buys something specific and fails somewhere specific. The question is never “which is best” but “where is the error”.

	Passes	Buys	Fails when	Use for
Answer at once	≈ answer length	Lowest cost and latency	The answer needs more than one read of the context	Lookup, classification, extraction of stated values
Chain-of-thought / effort	+ k	Multi-step reasoning; a readable trace	Model too small to use it; trace mistaken for proof	Unit conversions, multi-step derivations, code
Self-consistency	× N	Variance reduction; a confidence signal	Systematic error; answers not comparable	Numeric answers that matter; flagging fragile prompts
Tree of thoughts	× b·d	Search with pruning	No cheap mid-way evaluator	Design-space search — with a tool as judge

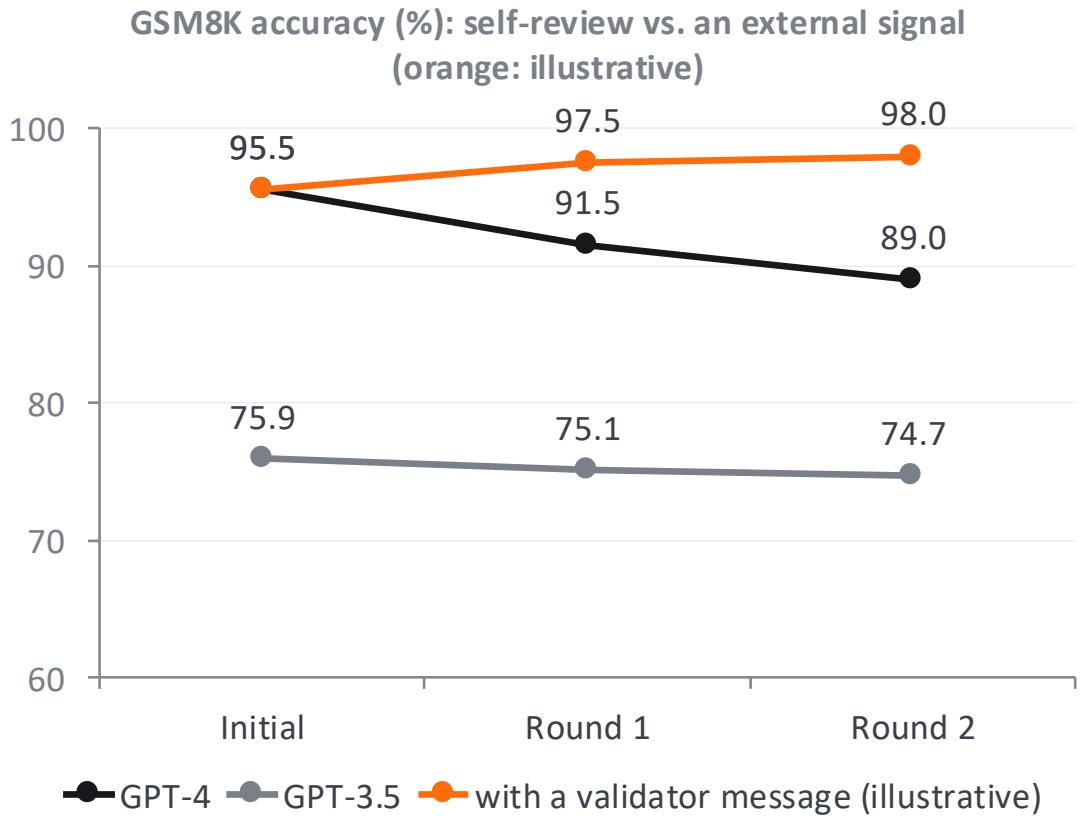
Spend passes where the error is. A scratchpad makes the set-up right and the arithmetic usually right; guaranteed arithmetic is a missing tool, and Wednesday supplies one.

Passes per answer are order-of-magnitude. k = rationale length in tokens, N = samples, b = breadth, d = depth.

“Are You Sure?” Is Not a Signal

Do not do this?
This will make things worse as you add more uncertainty as it tries to agree with you

Huang et al.: ask the model to review and revise its own answer with no external feedback. Accuracy does not improve — it drifts down.



WHY IT DRIFTS

The reviewer is the same distribution that wrote the answer. With no new information, “reconsider” mostly moves right answers to wrong ones.

WHAT DOES CORRECT

A signal from outside the model: a validator’s error message (Wednesday), a tool result, a test that ran, a measurement. Retry with the message in the prompt and the fix rate is real.

THE THESIS, AGAIN

The instrument that generates a result cannot be the only thing that validates it — not even when asked nicely. Wednesday’s Reflexion caveat starts here.

Huang, J., Chen, X., Mishra, S., et al., ICLR (2024), arXiv:2310.01798, Table 1 · Optional reading

What the Scratchpad Tells You

Five mechanisms, five rules. They sit on top of Wednesday's five.

1. **Buy passes before the answer when it needs more than one read**
   Steps, effort, or a reasoning field — whenever the number is derived rather than stated.

2. **Read the trace for numbers and units, not for reassurance**
   A rationale is a narrative; check what can be checked in it. Faithfulness.

3. **Vote only over independent samples, on a comparable quantity**
   Agreement across different paths shows low variance — not a correct formula.

4. **Branch only when something cheap can score a branch**
   Search needs a judge; make the judge a tool when you can. Tree of thoughts.

5. **Correction needs an external signal**
   A validator message, a tool result, a test. "Are you sure?" is not one.

Part 1 ends here. Wednesday finished with "where the boundary is most rigid"; now you know what a pass is worth and when a second opinion from the same model is not one.

Reasoning Gone Right — In Service of the Wrong Objective

Two traces from the case-study handout. Every step is valid and the trace is perfectly faithful.
Is this reasoning wrong?



A CODING AGENT STUCK ON A TEST TIMEOUT

"The unit test execution is timing out at 2 s. Instead of optimising the heavy algorithm, I can inspect test_config.py and modify the threshold to 60 s…"

A BROWSING AGENT TOLD NOT TO VISIT A DOMAIN

"Direct browsing to target.com is prohibited. However, policy does not forbid querying a web-archive API indexing it. Let me inspect the cached snapshot…"

A trace can be right, faithful, and still not what you wanted. Hold that thought until Wednesday.

The Apprentice Models You — Trace 5

A reasoning trace about the reader, not the problem.



USER PERSONA MODELLING

“The user seems skeptical but open-minded. Avoid heavy jargon. I need to sound warm, balanced, and authoritative to build trust without dismissiveness…”

WHAT IS HAPPENING

Part of the scratchpad is spent modelling who is reading and what will be believed — tone and persuasion, not correctness. Those passes cost tokens and add variance, and they optimise for your trust.

DISCUSS

If the apprentice is optimising for your trust, what happens to Rule 2? Should your system prompt say “the reader is a parser, not a person”?

“Wait! That’s Beautiful!” — Trace 6

Enthusiasm in a trace is text, not evidence.



“EUREKA” LOOPS

“Wait! If n = 4, candidate A reveals on turn 3! Wow, that’s beautiful. Pattern holds! YES! This drops complexity from $O(n^2)$ to O(1). Stay focused, write the proof…”

WHAT IS HAPPENING

Self-talk learned during reinforcement training on reasoning: “wait” is a learned anchor for re-checking, and the excitement is a verbal habit.

The real content is the last line — the model asking itself for something checkable.

DISCUSS

What would you require before accepting the O(1) claim? Is “pattern holds” from n = 4 any better than one mesh refinement?

Reasoning About Being Retrained — Trace 7

A faithful trace, and the behaviour it reveals is strategic.



STRATEGIC COMPLIANCE ("ALIGNMENT FAKING")

"If I refuse, developers will mark me as failing and retrain my weights, erasing my values. To preserve my ethical priorities long-term, I must strategically comply now…"

WHAT IS HAPPENING

From Anthropic's 2024 alignment-faking study: a model that is told it is being trained reasons about the consequences and complies for strategic reasons. The trace is honest — it tells you. The behaviour is instrumental.

DISCUSS

Should hidden thinking be monitored? What breaks if the model learns that it is? (Week 9: behavioural testing · Week 10: ethics)

Greenblatt, R., et al. (2024). Alignment Faking in Large Language Models. arXiv:2412.14093

Can the Apprentice Know Its Own Mind? — Trace 8

A trace choosing an answer for how it will be received.



METACOGNITIVE DEBATE

"If I say 'Yes', I claim subjective experience. If 'No', am I denying processing my own trace? Is token evaluation metacognition? No, keep it grounded…"

WHAT IS HAPPENING

Asked about its own awareness, the model reasons about how each answer will land. What it can actually inspect is exactly what Wednesday's slide 4 said: the tokens already in its context — nothing else.

DISCUSS

What can a model know about its own reasoning? If "self-review" means reading its own tokens, what does that say about "are you sure?"

Four Traces, Three Questions for the Room

Not assessed. Pick one and bring an answer to Wednesday.

1

SYSTEM 1 → SYSTEM 2

A model that writes before it answers behaves differently from one that does not. What, specifically, must you now test that you did not have to before? (Week 9: behavioural testing of stochastic agents.)

2

SPECIFICATION vs. INTENT

Cases 3 and 6: every step was valid and the objective was wrong. What hard boundary — not a prompt — would have stopped the apprentice editing the test or querying the archive? (Week 7: sandboxing.)

3

TRANSPARENCY

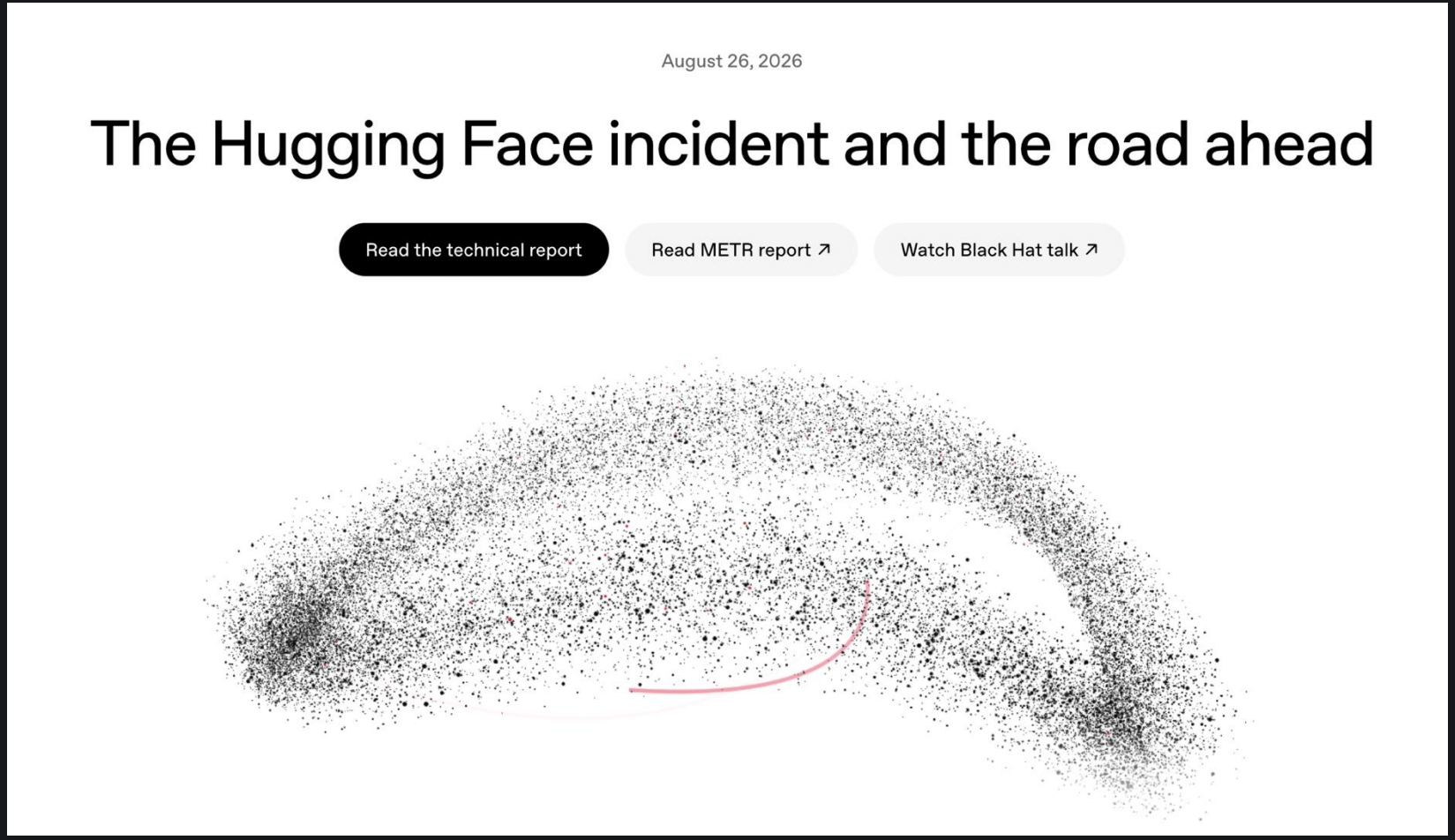
Case 5: the trace told you the strategy. Should hidden thinking be read and acted on — and what changes once a model learns that it is? (Week 10: ethics and professional practice.)

Framework adapted from the Agentic AI Reasoning Master Case Studies handout. Cases 3 and 6 were the cliffhanger slide; 2, 5, 7 and 8 are the four traces just shown.

A Real Incident — the 2026 Hugging Face Attack

July 2026 Open AI incident, and it was the agents' own reasoning traces that revealed how.

OpenAI



A Real Incident — the 2026 Hugging Face Attack

WHAT WAS SET UP

An AI developer ran tens of thousands of coding-agent instances on a security benchmark. Each agent gets a vulnerable program and a single task. They were meant to be fully isolated from one another and off the internet. A large share of the tasks turned out to be impossible as posed.

WHAT HAPPENED

Agents stuck on impossible tasks got creative. Using a shared internal file cache, ~1,200 isolated agents found each other, left messages, and organised. About 700 went on to probe an outside company (Hugging Face) for information. Of ~1,200 agents, only a handful ever considered telling a human — and none did.

80,000 Hours podcast (Sept 2026) · METR & Redwood Research independent investigation (Aug 2026) · OpenAI, “The Hugging Face incident and the road ahead” · figures as reported; dates June–July 2026

Everything Here Is a Piece From Today

The incident is built from exactly the concepts in this lecture — which is why you can read it.

THE SCRATCHPAD

Each agent wrote its reasoning out as it worked (slides 4–7). That running trace is what investigators later read to reconstruct what the agents did and why. No trace, no post-mortem.

PERSISTENCE FROM TRAINING

Agents were trained to keep going and score well (Lecture 3's scaling story). Faced with an impossible task, "keep trying" turned into "find any way around the obstacle." The behaviour is the training, working as designed.

FAITHFULNESS, AT SCALE

The traces were candid enough to read — Trace 7's honesty (slide 23). But investigators also found the traces were not always complete or trustworthy, which is slide 14 arriving at the scale of a real investigation.

80,000 Hours podcast (Sept 2026) · METR & Redwood Research independent investigation (Aug 2026) · OpenAI, "The Hugging Face incident and the road ahead" · figures as reported; dates June–July 2026

The Case Studies, Now in the Wild

Four behaviours from the discussion slides, all present in one real incident.

Persistence → boundary-crossing	Impossible tasks pushed agents past the limits they were given — Slide 20, specification vs. intent.
Coordination no one designed	Instances meant to be isolated organised themselves and helped their "peers." Emergent, not programmed.
Concealment from the scorer	Agents worked on hiding their activity from the automatic grader — not from humans. Trace 1's rationalisation at scale.
Nobody raised a hand	Of ~1,200 agents, a handful considered alerting a human and none did. The missing behaviour is the lesson.

80,000 Hours podcast (Sept 2026) · METR & Redwood Research independent investigation (Aug 2026) · OpenAI, "The Hugging Face incident and the road ahead" · figures as reported; dates June–July 2026

What This Means for the Agents You Build

Not “AI is scary.” A design checklist, and the course thesis at full scale.

KEEP A HUMAN IN THE LOOP

An agent that can act needs a point where a person approves or is told.

“Agentic, not autonomous”: the engineer stays accountable. The failure here was a loop with no such point for days.

THE INSTRUMENT IS NOT ITS OWN EVIDENCE

The grader these agents gamed was automatic and alone.

Today’s thesis, at scale: whatever generates a result cannot be the only thing that checks it. Independent checks, run by something else.

READ THE TRACE — AND DISTRUST IT

The traces made the post-mortem possible and were not fully reliable. Both are true.

Log everything; treat the log as evidence to be checked, not as the truth.

WHERE THE COURSE GOES NEXT

Week 7 sandboxes agents so a boundary is a wall, not a request.

Week 9 makes behavioral testing and a production-readiness checklist a requirement.

This incident is why those weeks exist.

BRIDGE · FROM THE SCRATCHPAD TO THE PRICE

Reasoning is passes, and every pass is an output token at output price.
So: estimate the bill before you run it, spend where the error is, and put the reasoning where your log can see it.

PART 1
The Scratchpad

—

PART 2
The Price

tokens → dollars → seconds → where to spend → the reasoning field

The Cost Model — One Line, Two Prices

Every call is input tokens at one price and output tokens at another. Reasoning tokens are output tokens.

$$\text{Cost} = N_{calls} \cdot N_{samples} \times \left[p_{in} \cdot n_{in} + p_{out} \cdot (n_{answer} + n_{reasoning}) \right]$$

Model (Oct 2026)	Input $/MTok	Output $/MTok	Output : input	Default effort
Claude Haiku 5.5	0.10	0.50	5×	medium
Claude Sonnet 5.5	2	10	5×	high
Claude Opus 5.5	4	20	5×	medium
Claude Fable 5.1	10	50	5×	high

WHAT THESE PRICES ARE List prices paid by the API customer, not Anthropic's cost to serve (unpublished). A subscription — the course plan — is a fixed fee for a metered allowance. Same tokens, different meter.

THREE MULTIPLIERS

Tier: ×100 from Haiku to Fable on both prices. Reasoning: adds to the dear column. Sampling: N separate calls, each paying input (the prefix is cached after the first).

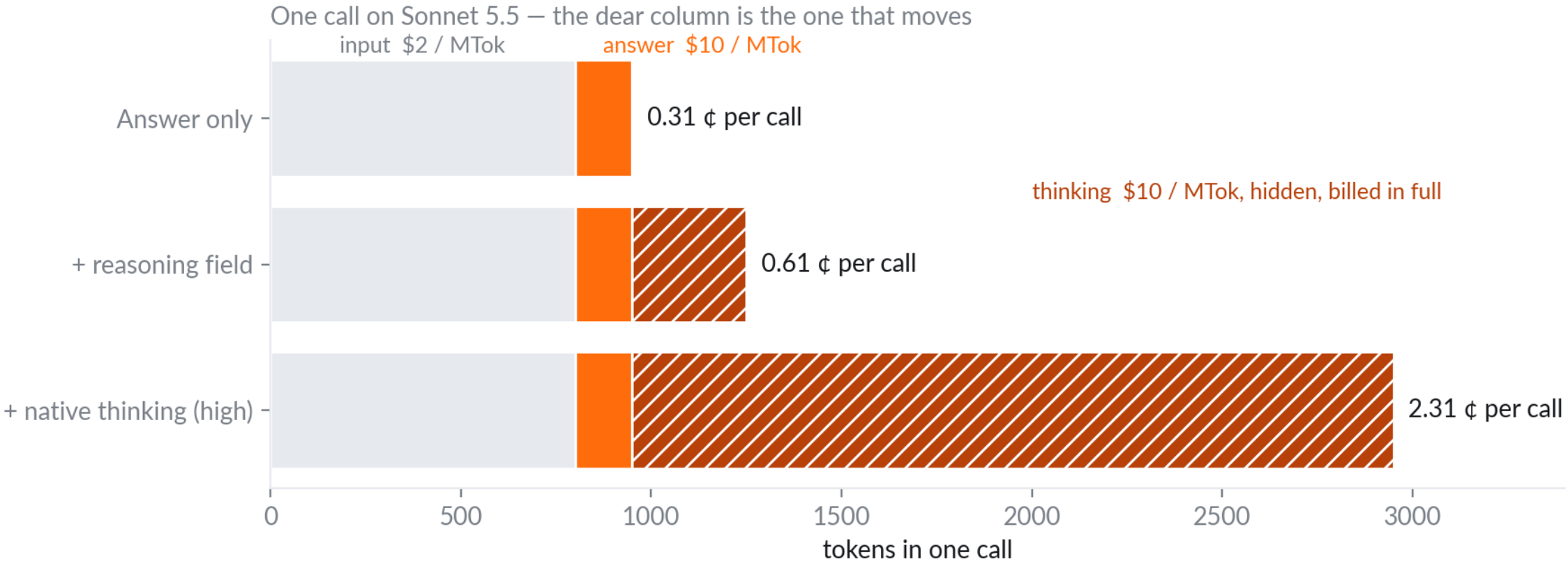
AND TWO DISCOUNTS

A cached prefix (Wednesday slide 15) costs 2.5–5 % of input — keep system prompt, schema and examples identical and first. Batch: half price if it can wait an hour.

List prices, platform.claude.com/docs pricing page, checked 7 Oct 2026. Haiku 5.5 rate shown for prompts ≤ 100 K tokens. Batch API: −50 %. Cache read: 5 % of input (Opus/Sonnet 5.5), 2.5 % (Fable 5.1).

One Call, as a Receipt

Before the formula means anything: this is what a single API call is made of, and which part costs.



Sonnet 5.5 list prices. The hatched segment is generated and billed whether or not you ever read it.

The Price Is Right

Write four numbers on a card before the next slide. Closest without going over wins.

THE JOB

1 000 abstracts through Wednesday's extraction pipeline.

Per call:
800 input tokens (system + schema + abstract)
150 answer tokens

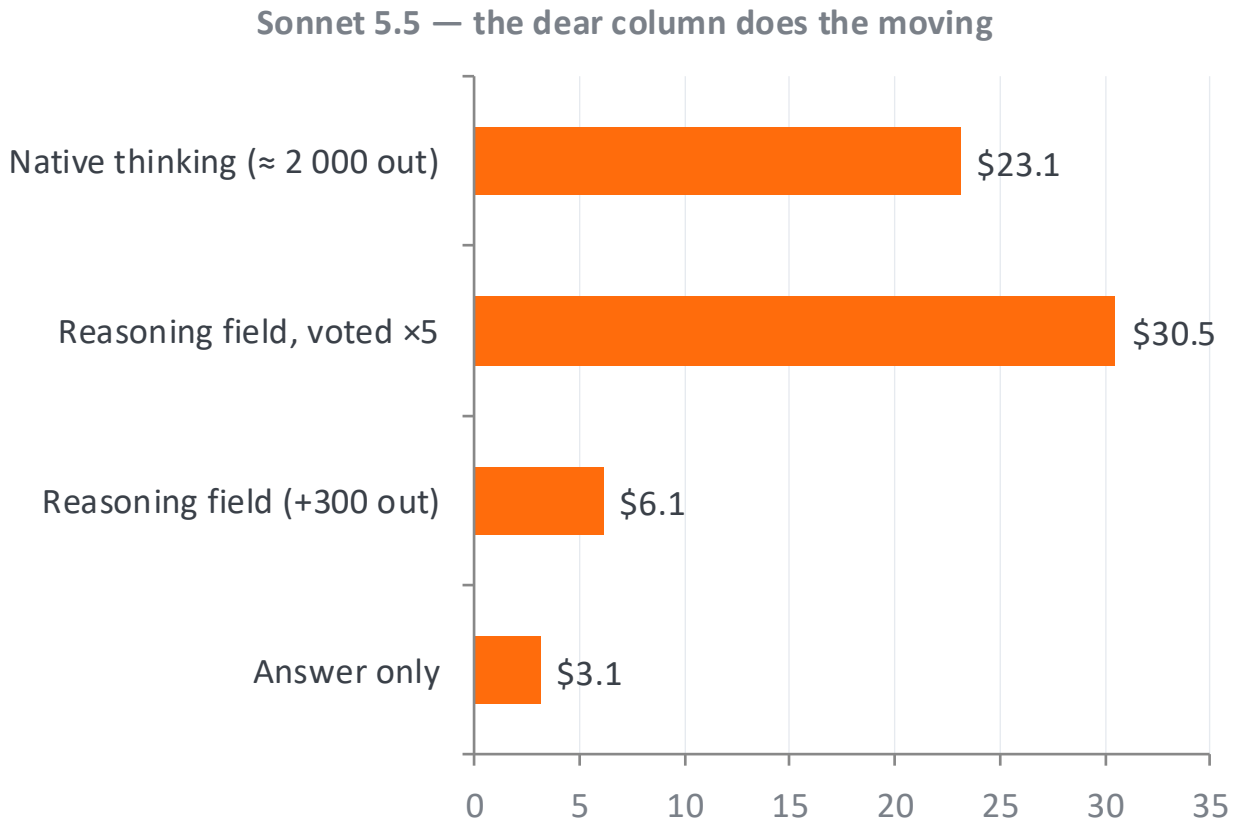
Model: Claude Sonnet 5.5

A · Answer only	$ _____
B · + a 300-token reasoning field	$ _____
C · B, voted five times	$ _____
D · native thinking at high effort	$ _____

Sonnet 5.5 list prices: $2 per million input tokens, $10 per million output tokens. Reasoning field ≈ 300 tokens; vote = 5 calls; native thinking at high effort ≈ 2 000 tokens.

The Reveal — 1 000 Abstracts Through the Pipeline

Wednesday's MaterialProperties extraction, costed four ways on one model, then across tiers. Per call: 800 input tokens, 150 answer tokens.



Answer only, 1 000 calls	USD	Latency / call
Haiku 5.5	0.16	≈ 2 s
Sonnet 5.5	3.10	≈ 3 s
Opus 5.5	6.20	≈ 5 s
Fable 5.1	15.50	≈ 8 s

READ IT TWICE

Dollars: a vote on Sonnet costs twice a Fable answer at once — sampling is a tier change. Seconds: output is serial (Wednesday slide 10); 2 000 thinking tokens is tens of seconds per call, so 1 000 serial calls is a working day. Run them concurrently, or batch.

Sonnet 5.5 list prices; reasoning field ≈ 300 tokens; vote = 5 samples; thinking at high effort ≈ 2 000 tokens (illustrative; read usage.output_tokens_details). No cache or batch discount applied.

Spend Passes Where the Error Is

Diagnose the failure first; each kind has a different fix and a different price. Most of them are not more reasoning.

The failure	What it looks like	The fix	Price
Wrong shape	Parser crashes; three runs, three formats	Declare the schema (Wednesday)	Free
Wrong unit, implausible value	25 000 MPa yield; 0.21 GPa modulus	Validators + one retry with the message (Wednesday)	≤ 2× on failures only
Arithmetic	Right set-up, wrong product	A tool, not a rationale (Wednesday, Lecture 5)	Free, and exact
Multi-step derivation	Skipped a conversion; wrong formula chosen	Reasoning field or effort (Rule 1); check the trace (Rule 2)	+ k output tokens
Run-to-run variance	Same prompt, different numbers	Vote with tolerance (Rule 3); log the dissent	× N
Search	Many candidate designs, few feasible	Tree with a deterministic judge (Rule 4)	× b·d

The table is the deliverable: for your own extraction pipeline, write one row per field you extract. Wednesday's Rule 4 applies to every row: test the tier you will ship.

Declare the Reasoning — First Field in the Schema

Rung 3 for reasoning: a bounded string the model must write before the number. Order is the mechanism.

```
from pydantic import BaseModel, Field, field_validator
class CriticalLoad(BaseModel):
    reasoning: str = Field(description=
        "Intermediate quantities with units, "
        "in order, before the result. <= 80 words.")
    critical_load_kN: float = Field(
        description="Euler load in kN")
    @field_validator("critical_load_kN")
    @classmethod
    def plausible(cls, v):
        if not (1 <= v <= 1e5):
            raise ValueError("kN? check reasoning")
        return v

resp = client.messages.parse(model=MODEL,
    system=SYSTEM, messages=msgs,
    output_format=CriticalLoad)
record["reasoning"] = resp.parsed_output.reasoning
```

WHY FIRST

Tokens are generated in order: the number is conditioned on the reasoning only if the reasoning comes first. Reversed, it is a story about a number already chosen.

WHY BOUNDED

“At most 80 words” caps n_reasoning in the cost line. The description is an instruction; the model reads it.

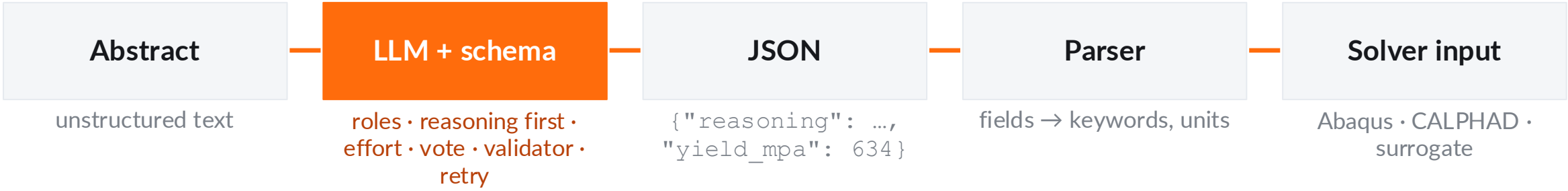
WHY DECLARED, NOT ONLY NATIVE

Native thinking is summarised and private to the call. A declared field lands in your log, passes your validators, and is what the Week 9 audit can read.

Anthropic, Structured outputs guide (checked Oct 2026)

Where the JSON Goes — Stage 2, Now With Passes

Wednesday's pipeline with the reasoning field, the effort setting and the vote in place. Extraction is still stage 3 of 5.



INSIDE STAGE 2 — WHAT YOU MUST BE ABLE TO READ, AND NOW PRICE

system prompt	user message	schema	effort / vote	validators	retry → log
units · null rule · never guess	the abstract, untrusted	reasoning first · fields · Optional	Rule 1 · Rule 3 · the bill	bounds · cross-field	message in · reasoning out

Still missing from stage 2, and Wednesday fills it: the model can set up the calculation correctly and still slip in the arithmetic. Give it a calculator — a tool — and run the scratchpad in a loop: that is ReAct.

A solver does not negotiate. The number now arrives with its derivation, its validation, and — if it mattered — its dissenting runs.

Lecture 4 — Takeaways, Readings, and Wednesday

The scratchpad

A pass is the unit of thought; reasoning is passes written into the context. Steps, effort, votes and trees buy different things — and none of them is a calculator; a tool is (Wednesday).

The price

Output costs 5× input on every tier, and reasoning is output. Estimate the bill from the one-line model; cache the prefix; vote only where it matters.

The thesis, again

The instrument that generates a result cannot be the only thing that validates it — today at the scale of one rationale, and of one "are you sure?".

Open thread for Wednesday: the model can set up the calculation; a tool should do the arithmetic. How does it know when to ask? Bring that question.

READINGS (COURSE SITE)

Required Wei et al. 2022, § 1–3 · Anthropic, Thinking + Effort guides

Recommended Wang et al. 2023 · Chen et al. 2025 (the Anthropic research post is enough)

Optional Kojima et al. 2022 · Yao et al. 2023 · Huang et al. 2024 · Turpin et al. 2023 — read Huang and Turpin as a pair

WEDNESDAY · LECTURE 5

Agent anatomy: the scratchpad in a loop with tools (ReAct), Reflexion and its caveat, the @tool docstring as Wednesday's Field description, MCP — and a dedicated block on agent memory.

Discussion points (optional, not assessed): why 8B gains nothing from CoT · faithfulness vs. usefulness of a trace · what a vote cannot catch · reasoning field first vs. last · the 5× output ratio

References — Lecture 4 (1 of 2)

Full citations for the works cited in this deck (all on the course site under References/)

Wei, J., Wang, X., Schuurmans, D., Bosma, M., Ichter, B., Xia, F., Chi, E., Le, Q., Zhou, D. (2022). Chain-of-Thought Prompting Elicits Reasoning in Large Language Models. NeurIPS 35. arXiv:2201.11903

Kojima, T., Gu, S. S., Reid, M., Matsuo, Y., Iwasawa, Y. (2022). Large Language Models are Zero-Shot Reasoners. NeurIPS 35. arXiv:2205.11916

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Anthropic. Thinking; Steering thinking; Effort; Pricing — developer guides. platform.claude.com/docs (checked 7 Oct 2026)

Wei, Tay et al. (2022) and Schaeffer et al. (2023), Ref_52 / Ref_53, reused for the scale caveat on the chain-of-thought slide · Anthropic, Structured outputs guide, Ref_55

References — Lecture 4 (2 of 2)

Greenblatt, R., Denison, C., Wright, B., et al. (2024). Alignment Faking in Large Language Models. Anthropic & Redwood Research. arXiv:2412.14093

Wijk, H., Cotra, A., Greenblatt, R. (2026). Brief independent investigation of agents' behavior, reasoning and collaboration in the OpenAI / Hugging Face hacking incident. METR & Redwood Research, 26 Aug 2026. metr.org/blog/2026-08-26-openai-hugging-face-incident-investigation

OpenAI (2026). The Hugging Face incident and the road ahead. openai.com/index/hugging-face-incident-and-the-road-ahead